

L2P Training Dataset — Construction, Curation & Quality

How we build the synthetic prompt → image dataset for the **L2P** run (converting a latent diffusion model to a pixel-space model), and the specific mechanisms that make it diverse, aesthetically strong, well-distributed, and clean.

1. What we're building & why

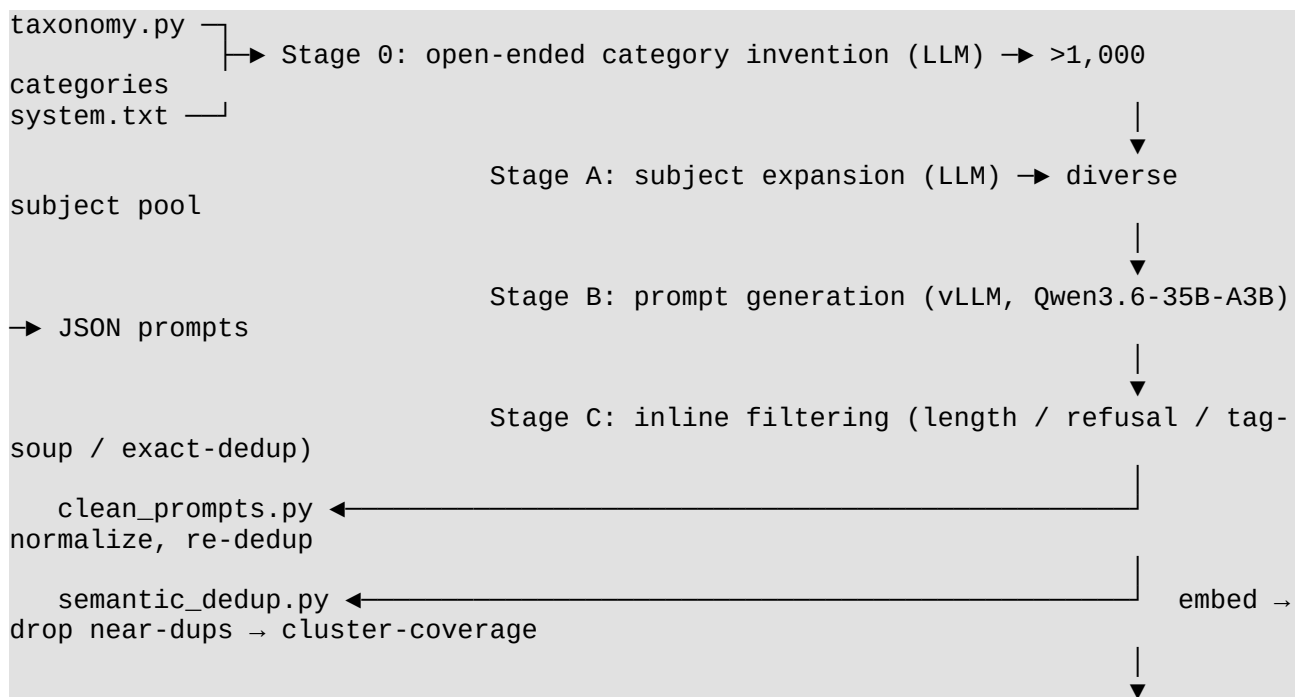
L2P ("*L2P: Unlocking Latent Potential for Pixel Generation*", arXiv:2605.12013) distills a pixel-space model from a **source LDM's own generations** — it never needs real data. So the dataset is two coupled artifacts:

1. **Prompts** — diverse, aesthetic, descriptive text prompts (this repo).
2. **Images** — generated from those prompts by the source model (**Qwen-Image-2512**), then aesthetically curated. These (prompt, image) pairs are the training corpus.

Sizing follows the paper: L2P used **~10k prompts** → **20k images** (2 seeds each) at 1024^2 , expanding 4 super-classes / 17 sub-classes into **>1,000 fine-grained categories**, with prompts concentrated at **200–350 chars**. We target the same shape (and overshoot to absorb filtering), so the final clean set is ~16–19k prompts.

Aesthetic north star: FLUX.1-Krea. "Make AI images that don't look AI." Quality over quantity, opinionated art direction, and engineered *against* the "AI look" (waxy skin, blurry backgrounds, washed-out/symmetric compositions, over-soft textures, palette collapse).

2. The pipeline (stage by stage)





3. How we ensure it's the *best* dataset

3.1 Diversity (cover the data manifold, no clustering)

- **Open-ended category invention** — instead of ~19 fixed buckets, an LLM invents >1,000 **fine-grained categories** (scaled as $\text{target}/8$), steered across photogenic themes per super-class. This is what the paper does and is the main diversity driver.
- **Per-batch randomization** — every generation batch samples a fresh combination of *viewpoint* $\times$ *lighting* $\times$ *style* $\times$ *color-grade* $\times$ *technique* (*film stock* / *lens* / *art movement*), plus rotated sentence openers and an explicit "don't reuse the previous prompt's words".
- **Measured result:** 13,196 distinct subjects (top subject 0.6%), opening-trigrams max **0.7%** (almost no template-iness), 169 distinct sub-classes.
- **Semantic dedup** (SemDeDup-style): embed every prompt (bge-small), drop near-duplicates at cosine > 0.90. Only ~1% **were near-dups** → the set was already highly diverse.
- **Coverage check:** k-means over the survivors; cluster-size **CV** $\approx$ **0.36–0.45** (even coverage, no dominant cluster).

3.2 Aesthetic skew (Krea-aligned)

- **system.txt** encodes an opinionated aesthetic: cinematic light, sophisticated color grading, film grain, tactile materials, asymmetric composition — and an explicit **anti-"AI-look"** block (no waxy skin, no blurred backgrounds, no washed-out palettes, no dead-centered symmetry).
- **Anti-crutch guards** — the data audit caught over-used words and we suppress them: "soft" (Krea explicitly flags over-soft textures), "ethereal/dreamy/glowing", and even a self-inflicted "tactile" (was hard-coded; now rotated across 8 phrasings).
- **Rich photographic vocabulary** injected per batch — film stocks (Portra, Ektar, CineStill, Velvia, HP5, Tri-X), lenses (35/85mm, anamorphic, tilt-shift, macro), color grades (teal-orange, bleach-bypass, technicolor, split-tone, kodachrome), art movements. Enrichment lifted rare-film vocab **0.1%** → **2.3%** and rare-grade **1.5%** → **7.3%**.
- **Image-level aesthetic selection (the biggest lever)** — `select_aesthetic.py` over-generates N seeds/prompt and keeps the top-k by **PickScore** (a *human-preference* model, deliberately **not** a LAION-aesthetic predictor, which Krea blames for the soft/symmetric AI look). This is the Krea "quality over quantity" step, enforced on pixels.

3.3 Distribution (deliberate, locked)

- **Aesthetic-weighted super-class mix: Nature 45 / Design 28 / People 22 / Synthetic 5.** Photogenic classes up-weighted; functional buckets (UI, slides, charts, raw text) kept minimal but present (~5%) so the model retains text-rendering ability.
- **Share-locking** — each category's weight = $\text{super_target} / (\#\text{categories in that super})$, so the prompt share is **exact regardless of invention noise** (verified: even when one super-class collapses to a few categories, shares stay at 45/28/22/5).
- **Length:** 250–450 chars (a touch above the paper's 200–350) for richer captions — dense captions are shown to improve T2I training. Measured median ≈ 306 , all in-band.

3.4 Quality (clean, safe, on-format)

- Inline Stage-C filtering: length window, refusal detection, "tag-soup" guard, exact dedup.
- `clean_prompts.py`: strips stray markdown/quotes/enumerators, collapses whitespace, re-dedups.
- **Audit leaks check:** 0 refusals, 2 JSON fragments, 19 non-Latin (text-render prompts) — clean.

3.5 Anatomy correctness (figure-study tier)

- A controlled tier (- - suggestive - frac, default 0.2, People-class only) injects tasteful, academic **figure-study / anatomy** framing — explicitly targeting what diffusion models get wrong: **hands, feet, joints, foreshortening, proportions, contrapposto**. Adults only, artistic/anatomical framing, no explicit sexual content. A dedicated People/anatomy augmentation batch is merged in so the data actually represents the human form well.

4. Cross-check vs. curation best practices

Best practice (lit. + Krea)	How we satisfy it
Exact + semantic dedup	<code>clean_prompts.py</code> (exact) + <code>semantic_dedup.py</code> (cosine>0.90)
Dense, descriptive captions	250–450 chars, median ~ 306 ; complete sentences, no tag-soup
Diversity / manifold coverage	>1,000 categories, 13k subjects, $CV \approx 0.4$ clusters, <1% near-dups
Aesthetic / preference filtering	PickScore top-k image selection (not LAION-aesthetic)
Balanced concept distribution	share-locked 45/28/22/5
Quality / safety filtering	inline + clean pass; 0 refusals, no leaks
Opinionated curation (Krea)	anti-AI-look system prompt, anti-crutch guards, Krea phrases

5. Image generation (Stage D) — settings

- **Source model:** Qwen-Image-2512 (bf16). **Attention:** FlashAttention-3 (Hopper) via HuggingFace kernels (kernels-community/flash-attn3) — verified, ~43% faster than SDPA.
 - **Recommended Qwen params:** `true_cfg_scale 4.0`, **28 steps**, **Euler scheduler** (chosen over DPM-Solver++ after a side-by-side — more natural faces/skin), official positive-magic suffix ("Ultra HD, 4K, cinematic composition.").
 - **Resolution:** L2P base is square **1024²**; Qwen-native is **1328²** with 7 aspect buckets (~1.6 MP). `--res-mode buckets` assigns content-aware aspect (landscapes wide, portraits tall, objects square).
 - **Steps:** 28 (DPM-Solver++) is the standard.
 - **FP8 tested** (`--fp8`, rowwise `torch._scaled_mm`): on H100 it gave only ~**10% speedup**
 - ~14% less memory with **comparable quality** (the real ~2× needs the fused v15 kernels, which are Blackwell-only). **bf16 kept as default**; FP8 available if memory-bound.
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6. Artifacts

file	what
<code>prompts.jsonl</code>	raw Stage-A → C output (~12k)
<code>prompts_clean.jsonl</code>	normalized + re-deduped
<code>prompts_aug.jsonl</code> / <code>prompts_anatomy.jsonl</code>	enriched + anatomy augmentations
<code>prompts_final.jsonl</code>	merged + semantically-deduped final prompt set
<code>images/ + manifest.jsonl</code>	generated images + (prompt, image, resolution) records

7. Reproduce

```
# prompts (enriched, aesthetic-weighted, anatomy-aware)
VLLM_USE_DEEP_GEMM=0 VLLM_MOE_USE_DEEP_GEMM=0 python generate_prompts.py --
target 12000 --output prompts.jsonl
python clean_prompts.py      --in prompts.jsonl      --out prompts_clean.jsonl
python semantic_dedup.py    --in prompts_clean.jsonl --out prompts_final.jsonl --
threshold 0.90

# images (FA3, buckets, Qwen-recommended params) + aesthetic top-k selection
```

```
python qwen-image-optimizations/src/batch_infer.py --prompts prompts_final.jsonl \
    --res-mode buckets --no-attn-mask --magic --batch 4 --seeds 4    # over-
generate
python select_aesthetic.py --prompts prompts_final.jsonl --seeds 5 --keep 2    #
PickScore curation
```

8. Next steps — AFTER generation finishes

Do these in order on the generated (prompt, image) set once `batch_infer` is done and the shards are on the Hub. **For the L2P transfer, prioritise coverage + failure-removal over peak aesthetics** (the transfer is distillation — it needs the teacher's manifold covered, not the prettiest 40%).

8.1 Curate — remove failures, keep good seeds (`curate_images.py`)

Per-image keep decision over the 2 seeds: both good → keep both, one good → keep one, both bad → drop both. **Be permissive for the transfer** — only cut clear failures (broken anatomy, artifacts, mode collapse), not merely "less pretty" images.

```
# 1) look at the score distribution first
python curate_images.py --repo-id <you>/l2p-dataset --stats-only
# 2) permissive keep for the TRANSFER set (keep most; drop only the bottom
failures)
python curate_images.py --repo-id <you>/l2p-dataset --keep-percentile 85 --keep
2
# (for the later aesthetic FINETUNE, re-curate the same shards strictly: --
keep-percentile 45)
```

- Keeping BOTH seeds when both are good is correct and encouraged for the transfer — two valid teacher samples per prompt = more manifold coverage. `--keep 2` does this.
- Whether to keep-both vs keep-one is decided per-image by the score floor, so you don't set a global "split"; you set the **floor** (start ~top-85% for transfer).

8.2 Diversity audit — measure VISUAL coverage, not noun count (`diversity_audit.py`)

Prompt-dedup measured text diversity; this measures the image manifold (the real "is it just shrimp/bear/ocean?" check) and emits per-image sampling weights.

```
python diversity_audit.py --repo-id <you>/l2p-dataset --k 200
# -> diversity_report.json (CV, effective #modes, dominant modes)
# -> sampling_weights.jsonl ({id, cluster, weight})
```

Health check: **CV < 0.5** and **largest visual mode < ~3–4%** = good coverage. If **top-10 modes > ~25%**, a few visual modes dominate — fix it at train time (next step), not by regenerating.

8.3 Apply inverse-frequency sampling in the transfer dataloader

Feed `sampling_weights.jsonl` to a `WeightedRandomSampler` (or multiply per-sample loss by weight). Over-represented visual modes get down-weighted so they don't dominate gradients — fixes the repetition concern **without deleting or regenerating anything**.

8.4 (optional) Image-level dedup

Prompt-dedup doesn't catch images from different prompts that *look* identical. If 8.2 flags heavy redundancy, run a pHash + embedding image dedup (same approach as `finetune_clean/pipeline/02_dedup.py`) before training.

8.5 Then: training staging

1. **L2P transfer** — this curated synthetic set (1024; downscale 1328 at load) + 8.3 weights.
2. **Base aesthetic finetune** — `finetune_clean/out/base/` (cleaned Pexels) + ~10–30% synthetic.
3. **4K finetune** — `finetune_clean/out/4k/` (UltraHR subset) + 4K-eligible Pexels people.

8.6 Human preference ranking — needed or not?

Not needed for the transfer. Distillation just needs failures removed and coverage kept; PickScore (8.1) does that. Exhaustively ranking all ~20k pairs with 3–4 annotators is high-cost, low-ROI here.

It becomes valuable **only if you later add a preference-optimisation stage** (DPO / reward-model / RLHF-style aesthetic alignment, the Krea-style "make it match human taste" step). Even then:

- you'd sample a **subset (~2–5k pairs)**, not rank everything;
- the right move now is a **small calibration set (~200–400 pairs)** hand-ranked by your reviewers, used to *check PickScore agrees with your taste*. If agreement is high, trust the automated scores for the bulk; if not, recalibrate the threshold/scorer. That buys ~95% of the benefit at ~1% of the labelling cost.